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$$f(x, y) = x^3 - x^2y + xy^2 - y^3 \text{ met } a(1, -1) \text{ en } h(2, 3)$$

$$Df(a, h) = \lim_{\lambda \rightarrow \infty} \frac{f(a + \lambda h) - f(a)}{\lambda}$$

$$f(a) = 4$$

$$f(a + \lambda h) = f(1 + 2\lambda, -1 + 3\lambda) = (1 + 2\lambda)^3 - (1 + 2\lambda)^2(-1 + 3\lambda) + (1 + 2\lambda)(-1 + 3\lambda)^2 - (-1 + 3\lambda)^3$$

$$= -13\lambda^3 + 28\lambda^2 - 6\lambda + 4$$

$$\Rightarrow \lim_{\lambda \rightarrow \infty} \frac{f(a + \lambda h) - f(a)}{\lambda} = \frac{-13\lambda^3 + 28\lambda^2 - 6\lambda + 4 - 4}{\lambda} = -6$$

References